

Weak Learning Failure Alone Is Not Gradient Starvation

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Working draft

Abstract

A weak predictive feature can remain poorly learned because its isolated learning is inherently slow, a stronger cue suppresses cross-entropy sample weights, or coupled parameter geometry penalizes its update. These explanations are not causally equivalent. We introduce a matched weak-only counterfactual: paired models share initialization, data sequence, weak coordinates, and noise; only the strong cue is ablated. Causal starvation requires both outcome suppression and a nondegenerate weak-only first hit; gate failure leaves causal status indeterminate. For full-batch logistic gradient flow, we derive exact finite-width dynamics, $\dot{M}_w = c_w^\top q(r)/n$ and $\dot{r} = \Theta q(r)/n$, and the pathwise identity $\Delta = P - N$. A general prospective theorem converts certified parameter-region bounds for any smooth finite network into finite-horizon predictions; a fully trained five-parameter tanh CNN supplies a certified nonlinear existence and no-suppression control pair. In a frozen matched CIFAR-10 study, all 8/8 fresh seed blocks pass both the causal certificate and its stricter at-hit subtype; the cue-trained model loses 19.9 percentage points on neutral inputs and 31.9 points under cue conflict. A common fresh linear head recovers part, but not all, of the neutral gap.

1 Introduction

When a deep neural network learns one dominant feature while neglecting another predictive cue, the neglected signal is frequently described as “starved” [Pezeshki et al., 2021]. However, observing a small response to a weak feature during training is not by itself causal evidence of gradient starvation: the weak feature may simply be unlearnable under the chosen architecture, optimization horizon, or signal-to-noise ratio. The distinction is acute in recurrent neural networks, where cue strength, temporal separation, parameter sharing, and non-static geometry jointly govern representation learning [Bengio et al., 1994, Hochreiter and Schmidhuber, 1997, Cho et al., 2014].

To isolate the causal effect of a strong cue, we introduce a matched counterfactual intervention grounded in formal causal inference [Pearl, 2009, Peters et al., 2017]. A both-feature model $\mathcal{M}_B$ is paired with a weak-only model $\mathcal{M}_W$ that shares identical weight initialization, sample sequence, weak inputs, and optimizer state; only the strong input channel is zeroed out. Tracking the paired weak-feature response gap $\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau)$ and its time derivative $d(\tau) = \dot{\Delta}(\tau)$, positive d indicates *relative transfer*, negative d indicates *rate suppression*, and negative Δ indicates *outcome suppression*. We reserve the term *causal starvation* strictly for settings where outcome suppression occurs alongside a successful weak-only first hit of a prespecified target β within horizon H . When the weak-only gate fails, causal status is indeterminate: neither starvation nor its absence is established.

Three core contributions.

1. **Causal diagnosis.** We formalize the matched weak-only counterfactual and a learnability-gated causal certificate, proving that weak learning failure alone is insufficient to diagnose starvation. The stricter at-hit endpoint is retained as an empirical robustness subtype.
2. **Finite-width mechanism and prospective prediction.** Exact response–logit flow and the pathwise identity $\Delta = P - N$ separate sample weighting, parameter geometry, rate suppression, and outcome suppression. A general smooth-network theorem turns independently certified parameter-region bounds into prospective finite-horizon predictions, with a certified fully trained tanh-CNN existence and control pair; no universal positive-lag phase boundary is claimed.

3. **Controlled evidence.** A frozen CIFAR-10 confirmation yields both certificates in 8/8 paired seeds, persistent core-response deficits, large neutral/conflict harm, and partial fresh-head recovery. Recurrent experiments show why the learnability gate matters. CDC supplies a supporting local mechanism-control result, not a broad optimizer-superiority claim.

2 Related work

Feature starvation and shortcut learning. Cross-entropy loss can favor rapidly learnable features strongly enough to suppress gradient signal to slower predictive features, an effect termed gradient starvation [Pezeshki et al., 2021]. Shortcut learning, simplicity bias, and early memorization describe the broader empirical tendency of deep networks to exploit low-complexity, unintended cues at the expense of robust core features [Arpit et al., 2017, Geirhos et al., 2020, Shah et al., 2020, Hermann and Lampinen, 2020, Scimeca et al., 2022]. Invariant risk minimization and failure-guided debiasing seek representations invariant to spurious correlations [Arjovsky et al., 2019, Nam et al., 2020]. Our contribution is diagnostic: the matched weak-only counterfactual explicitly tests whether the neglected feature would have been learned in the absence of the strong cue, while direct drift and pathwise attribution distinguish transient rate suppression from adverse outcome suppression.

Methods such as Group DRO, Just Train Twice, and last-layer retraining target worst-group performance under spurious correlations [Sagawa et al., 2020, Liu et al., 2021b, Kirichenko et al., 2023]. We address a distinct question—causal attribution of feature-level learning dynamics—and do not propose a competing robustness benchmark. Our semi-real study evaluates controlled cue ablation on standard image benchmarks [LeCun et al., 1998, Xiao et al., 2017], preserving real core pixels while isolating causal feature interactions.

Implicit bias and margin dynamics. On separable classification problems trained with exponential or logistic losses, gradient flow exhibits an implicit bias toward directional maximum-margin separators in linear and homogeneous models [Soudry et al., 2018, Gunasekar et al., 2018, Ji and Telgarsky, 2019]. While asymptotic margin maximization governs directional convergence at infinite time, gradient starvation is an optimization-time transient phenomenon where rapid growth of dominant logit responses diminishes sample weights before secondary features can develop. Our exact finite-width equations capture this trajectory-level competition directly.

Wide and recurrent learning dynamics. Neural tangent kernels (NTK) characterize infinite-width gradient flow in the lazy training regime [Jacot et al., 2018, Du et al., 2019, Arora et al., 2019, Chizat et al., 2019], and wide-network linearization motivates frozen-tangent approximations [Lee et al., 2019]. Beyond the lazy regime, feature learning limits describe representation movement in overparameterized models [Yang and Hu, 2021, Bordelon and Pehlevan, 2022]. Our response-logit equations are instead exact at finite width for arbitrary trajectories. Freezing their kernels defines a nonlinear logistic tangent surrogate, but transferring its predictions to a trained network requires kernel-movement bounds that we do not establish for tanh RNNs or GRUs. Low-dimensional overlap theories for low-rank RNNs demonstrate that loss-invisible modes retain training history [Ger and Barak, 2026], echoing classical insights on disordered recurrent dynamics [Sompolinsky et al., 1988] and cautioning against closure in only the visible response coordinates. Mean-field descriptions of structure and disorder in trained recurrent circuits [Clark et al., 2026] study a distinct long-time equilibrium setting. We therefore do not claim a general recurrent optimization-time mean-field theory.

Gradient surgery and multi-task optimization. Multi-task learning often suffers from destructive gradient interference, inspiring multi-objective formulations including MGDA [Sener and Koltun, 2018], conflict-averse descent (CAGrad) [Liu et al., 2021a], gradient sign dropout (GradDrop) [Chen et al., 2020], rotational alignment (Rotograd) [Javaloy and Valera, 2022], and bargaining game approaches [Navon et al., 2022]. PCGrad projects conflicting task gradients onto orthogonal subspaces [Yu et al., 2020]; Bloop stabilizes auxiliary projections using an exponential moving average [Hsieh et al., 2024]. CDC also employs a projection, but its protected direction is the gradient of a strong-feature response and its target is derived from the matched weak-only trajectory. That extra counterfactual requires a constructible ablation, a second model,

Table 1: Mathematical terminology and causal interpretation in the paired diagnostic framework.

Quantity	Sign	Causal / Dynamical Interpretation
$d(\tau) = \dot{\Delta}(\tau)$	> 0	Relative transfer: strong cue accelerates instantaneous weak learning.
$d(\tau) = \dot{\Delta}(\tau)$	< 0	Rate suppression: strong cue decelerates instantaneous weak learning.
$\Delta(\tau)$	< 0	Outcome suppression: accumulated weak response is worse under both cues.
$\tau_W^*(\beta)$	$\in (0, H]$	Weak-only learnable: weak-only control achieves target β within horizon H .
$\tau_W^*(\beta)$	$> H / \infty$	Weak-only unlearnable: gate fails; causal status is <i>indeterminate</i> .

lockstep training, and full-batch gradients. Our comparisons use adapted shadow-target variants and support neither equivalence to the original methods nor broad CDC superiority.

3 Matched causal diagnosis

3.1 The matched weak-only counterfactual

Let condition $C \in \{\text{B}, \text{W}\}$ denote the both-feature and weak-only training runs, respectively. The two models share an identical parameter initialization $\theta_{\text{B}}(0) = \theta_{\text{W}}(0) = \theta_0 \in \mathbb{R}^P$, identical training sequence samples $(x_i, y_i)_{i=1}^n$, and identical optimization hyperparameters. The only difference is an intervention on the input: in W, the strong cue channel is set identically to zero ($x_{s,i} \equiv 0$).

Let $M_s^C(\theta), M_w^C(\theta)$ be differentiable scalar functionals measuring the model’s response to isolated strong and weak cues. For linear models, these correspond to channel-localized mode projections; for nonlinear recurrent models, they are defined via odd symmetric unit-probe contrasts: $M_a^C(\theta) = [f_C(x_a^+; \theta) - f_C(x_a^-; \theta)]/2$. For logistic gradient flow on full-batch signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$, the empirical loss is:

$$L_C(\theta) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-r_i^C(\theta)} \right), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (1)$$

The paired response gap $\Delta(\tau)$ and instantaneous equal-time relative drift $d(\tau)$ are:

$$\Delta(\tau) = M_w^{\text{B}}(\tau) - M_w^{\text{W}}(\tau), \quad d(\tau) = \dot{\Delta}(\tau) = \dot{M}_w^{\text{B}}(\tau) - \dot{M}_w^{\text{W}}(\tau). \quad (2)$$

Since $\theta_{\text{B}}(0) = \theta_{\text{W}}(0)$, the gap begins at zero: $\Delta(0) = 0$.

3.2 Transfer, rate suppression, and outcome suppression

Table 1 summarizes the formal interpretation of signs in the paired framework.

3.3 Formal causal-starvation certificate

We now state the central definition of the paper.

Definition 1 (Causal starvation certificate). Fix a target response threshold $\beta > 0$ and optimization horizon $H > 0$. Let $\tau_W^*(\beta) = \inf\{\tau \in [0, H] : M_w^{\text{W}}(\tau) \geq \beta\}$ denote the first hitting time of the weak-only control, with $\inf \emptyset = \infty$. A training run exhibits a **causal starvation certificate** if and only if:

1. **Nondegenerate weak-only learnability:** $\tau_W^*(\beta) \in (0, H]$ with $\beta > M_w^{\text{W}}(0)$; and
2. **Outcome suppression:** there exists $t_s \in (0, H]$ such that $\Delta(t_s) < 0$.

The run also satisfies the stricter **at-hit starvation** subtype when $\Delta(\tau_W^*(\beta)) < 0$. If $\tau_W^*(\beta) > H$, the weak-only gate fails and the causal status is **indeterminate**. If $\beta \leq M_w^{\text{W}}(0)$, the target is met at initialization (degenerate) and licenses no causal claim.

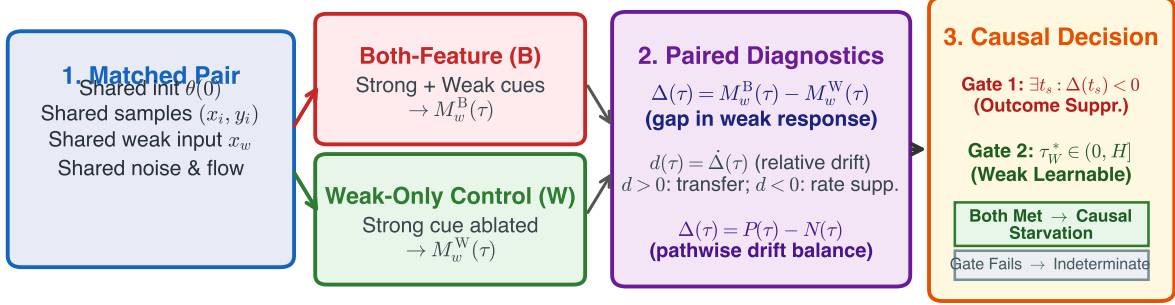


Figure 1: Schematic of the matched counterfactual causal diagnosis. A both-feature model $\mathcal{M}_B$ and weak-only control $\mathcal{M}_W$ share initialization $\theta(0)$, data samples (x_i, y_i) , and noise. The strong cue is ablated in W. Causal starvation is certified if and only if both outcome suppression ($\Delta < 0$) and the weak-only learnability gate ($\tau_W^* \leq H$) are satisfied.

4 Exact finite-width dynamics and pathwise attribution

4.1 Exact finite-width response–logit dynamics

We establish that the coupling between sample weighting and parameter geometry is an exact finite-width chain-rule identity rather than an asymptotic mean-field closure.

Proposition 1 (Exact finite-width response–logit dynamics). *Under full-batch logistic gradient flow (Eq. 1) at any finite width and parameter dimension P , let $J^C(\tau) \in \mathbb{R}^{n \times P}$ have rows $\nabla r_i^C(\theta_C(\tau))^\top$. Define the empirical tangent kernel $\Theta^C(\tau) = J^C(\tau)J^C(\tau)^\top \in \mathbb{R}^{n \times n}$, response-coupling vector $c_w^C(\tau) = J^C(\tau)\nabla M_w^C(\theta_C(\tau)) \in \mathbb{R}^n$, and sigmoid sample-weight vector $q(r^C)_i = \sigma(-r_i^C)$. Then:*

$$\dot{M}_w^C(\tau) = \frac{1}{n} c_w^C(\tau)^\top q(r^C(\tau)), \quad \dot{r}^C(\tau) = \frac{1}{n} \Theta^C(\tau) q(r^C(\tau)). \quad (3)$$

Proof Sketch. By chain rule, $\dot{\theta}_C = -\nabla L_C = \frac{1}{n} \sum_i \sigma(-r_i^C) \nabla r_i^C = \frac{1}{n} (J^C)^\top q(r^C)$. Differentiating M_w^C and r_i^C along $\dot{\theta}_C$ immediately gives Eq. (3). Full proof in Appendix C.1. $\square$

Interpretation. Eq. (3) factorizes learning velocity into two distinct terms:

- $q(r)$: The cross-entropy weighting vector. As logits r_i grow, $q_i \rightarrow 0$, diminishing the effective learning rate.
- c_w : The response geometry vector, reflecting alignment between gradients of M_w and sample logits r_i .

Corollary 1 (Closed projected flow under exact realization). *If signed sample logits admit the exact parameter-independent decomposition $r_i^C = \sum_b z_{ib}^C M_b^C + r_i^\perp$, then:*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C + R_a^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n} \sum_i z_{ib}^C \sigma(-r_i^C), \quad (4)$$

where $R_a^C = \frac{1}{n} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle$. When realization is exact ($r^\perp \equiv 0$), $R_a^C \equiv 0$ and the response dynamics close autonomously.

4.2 Cumulative drift balance

Proposition 2 (Cumulative drift balance). *Let $\Delta(\tau)$ be absolutely continuous on $[0, H]$ with $\Delta(0) = 0$ and $d(\tau) = \dot{\Delta}(\tau)$ almost everywhere. Define positive and negative drift components:*

$$P(t) = \int_0^t [d(u)]_+ du, \quad N(t) = \int_0^t [-d(u)]_+ du. \quad (5)$$

Then for all $t \in [0, H]$:

$$\Delta(t) = P(t) - N(t), \quad \text{and} \quad \Delta(t) < 0 \iff N(t) > P(t). \quad (6)$$

Proof. Since $\Delta(t) = \Delta(0) + \int_0^t d(u) du = \int_0^t ([d(u)]_+ - [-d(u)]_+) du = P(t) - N(t)$. Full details in Appendix C.2. $\square$

Core implication. A negative relative rate ($d(t) < 0$) does *not* imply outcome suppression ($\Delta(t) < 0$) unless the accumulated negative drift $N(t)$ repays all accumulated positive transfer $P(t)$.

Corollary 2 (Single-crossing tail-area criterion). *Suppose there exists $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ on (τ_d, H) . Let $A_+ = P(\tau_d) = \Delta(\tau_d) > 0$. Then for $t \geq \tau_d$, $\Delta(t) = A_+ - N(t)$. Strict outcome suppression occurs by horizon H if and only if $N(H) > A_+$.*

5 General prospective finite-horizon prediction

Let $\theta_C \in \mathbb{R}^P$ contain all trainable parameters of condition C , with $L_C, M_w \in C^2$, $V_C = -\nabla L_C$, and $X = (\theta_B, \theta_W)$. Define $V = (V_B, V_W)$, $\Delta(X) = M_w(\theta_B) - M_w(\theta_W)$, $d = \mathcal{L}_V \Delta$, and $j = \mathcal{L}_V d$. These are exact full-parameter quantities; no tangent freezing or response decomposition is assumed.

Theorem 1 (Prospective smooth-network certificate). *Fix $0 \leq t_w < H$, $h = H - t_w$, and a validated enclosure of $X(t_w)$ whose points lie within r_w of midpoint $\bar{X}_w$. On $U_R = \{X : \|X - \bar{X}_w\| \leq R\}$ suppose uniformly that*

$$\|V(X)\| \leq S, \quad j(X) \leq -\lambda < 0, \quad \dot{M}_w^W(X) \geq \nu > 0, \quad r_w + Sh \leq R. \quad (7)$$

If the enclosure certifies $\Delta(t_w) \in [D_-, D_+]$, $d(t_w) \in [d_-, d_+]$, and $M_w^W(t_w) \geq m$, then for $0 \leq u \leq h$,

$$\Delta(t_w + u) \leq Q(u) := D_+ + d_+ u - \frac{\lambda u^2}{2}, \quad M_w^W(t_w + u) \geq m + \nu u. \quad (8)$$

If some prespecified $u_a \in (0, h)$ satisfies $Q(u_a) < 0$ and $d_+ - \lambda u_a \leq 0$, then suppression persists on $[t_w + u_a, H]$. If also $\beta > \max\{0, M_w^W(0)\}$ and $m + \nu h \geq \beta$, the weak-only gate passes and the run satisfies Definition 1. If the weak-only first hit is independently certified to lie in $[t_w + u_a, H]$, the conclusion strengthens to the at-hit subtype. Conversely, under the same containment, $|j| \leq J$, $D_- \geq 0$, and $d_- - Jh > 0$ exclude suppression after t_w .

The result applies to any finite smooth computational graph for which the uniform premises are certified before observing the future trajectory. It does not assert that those premises hold for typical networks or initializations. The full proof is in Appendix C.5.

Corollary 3 (Certified nonlinear tanh-CNN existence and control). *Consider the fully trained one-dimensional, two-input-channel shared-kernel network*

$$f_\theta(x_s, x_w) = \frac{c}{2} \sum_{p=0}^1 \tanh(s_0 x_{s,p} + s_1 x_{s,p+1} + w_0 x_{w,p} + w_1 x_{w,p+1}), \quad (9)$$

with $\theta = (s_0, s_1, w_0, w_1, c)$. For $y \in \{-1, 1\}$, B uses $x_s = (y, y, 0)$ and $x_w = (0, y, y)$, while W sets $x_s = 0$ and retains x_w . Let the common weak probe be the odd contrast at $x_s = 0$ and $x_w = \pm(0, 1, 1)$. Under mean-logistic

gradient flow from $\theta_0 = (-10^{-4}, -5 \times 10^{-5}, 0, 0, 1)$, with $H = 1/250$ and $\beta = 3/2000$, an outward interval certificate proves

$$\begin{aligned} \Delta(t) &> 0 & (0 < t \leq 10^{-4}), \\ \Delta(t) &\leq -10^{-7} & (3/1000 \leq t \leq 1/250), \\ 0 &< \tau_W^*(\beta) &\leq 149/50000. \end{aligned} \tag{10}$$

Thus this exact nonlinear network exhibits transfer followed by a broad causal-starvation certificate. Exact-real full-batch gradient descent with step size 10^{-4} likewise certifies $\Delta_k \leq -10^{-7}$ for updates 30–40 and the W gate by update 40. For the same architecture initialized at $(-3/20, -3/40, 0, 0, 1)$, a separate certificate proves $\Delta(t) > 0$ after warm-up through $H = 1/500$, giving a matched no-suppression control.

The construction proves nonvacuity for a smooth nonlinear trained network, not prevalence under random initialization, practical-scale harm, or the stricter at-hit subtype. Exact logits, interval bounds, and certificate arithmetic are in Appendix C.6.

5.1 Exact dense-RNN initialization special case

For the noiseless rank-one dense linear RNN ($h_{k+1} = Wh_k + Bx_k$, $f(x) = c^\top h_T$), the initial drift needed by the general theorem is available exactly.

Theorem 2 (Realized-initialization rate-gap criterion). *Let the strong cue have strength $\rho > 0$ at lag 0, and the weak cue have lag $\ell \geq 0$. Define $K = G_{ws}(0) = b_s^\top W^\ell b_w$, $A = G_{ww}(0) = \|W^\ell b_w\|^2 + \|(W^\ell)^\top c\|^2 + \|S_w\|_F^2$, $s_B = \sigma(-[\rho M_s(0) + M_w(0)])$, and $s_W = \sigma(-M_w(0))$. The initial rate gap is:*

$$d_0 = s_B(\rho K + A) - s_W A, \quad d_0 > 0 \iff K > \frac{A}{\rho} \left(\frac{s_W}{s_B} - 1 \right). \tag{11}$$

Since $\Delta(0) = 0$, $d_0 < 0 \implies \Delta(t) < 0$ for sufficiently small $t > 0$.

This criterion is exact at $t = 0$ but local. It supplies a realized-parameter input to Theorem 1, not a distribution-only positive-lag phase boundary.

6 Counterfactual Drift Correction

To evaluate whether the diagnosed mechanism can be controlled locally, we introduce Counterfactual Drift Correction (CDC). In a fixed Euclidean parameter chart, let $a = \nabla m_s(\theta)$, $p = \nabla m_w(\theta)$, and $v = -\nabla L(\theta)$. Project p onto the orthogonal complement of a :

$$q = p - \frac{p \cdot a}{\|a\|^2} a \quad (a \neq 0), \quad \delta = [F_w^W - p \cdot v]_+, \quad \alpha = \frac{\delta}{\|q\|^2} \quad (q \neq 0). \tag{12}$$

The corrected parameter update is $v' = v + \alpha q$.

Theorem 3 (CDC local guarantee). *In the fixed parameter chart:*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly (strong drift is unaffected).
2. **Minimum Euclidean norm:** $d^* = \alpha q$ is the unique minimum-norm correction attaining the weak-drift lower bound $p \cdot (v + d) \geq F_w^W$ subject to $a \cdot d = 0$.
3. **Local finite-step deviation:** For L -Lipschitz ∇m_s , $|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2) = O(\eta^2)$.

CDC provides first-order instantaneous mechanism control; it is not a claim of global optimization superiority.

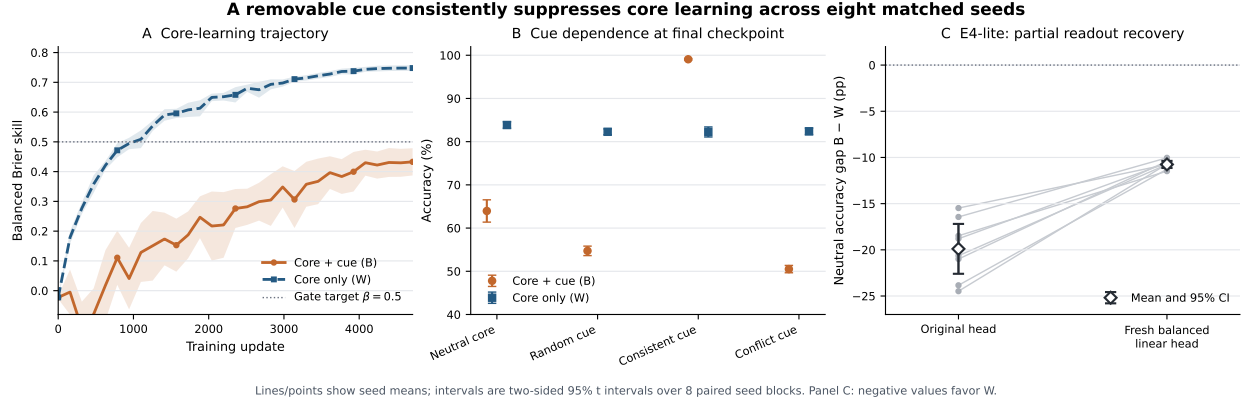


Figure 2: Frozen matched CIFAR-10 confirmation across eight paired seed blocks. Lines and points are seed means with two-sided 95% t intervals. B contains the removable cue; W contains the identical core with a neutral cue channel. Every seed passes both the causal certificate and at-hit subtype. The fresh-head test recovers part, but not all, of the neutral gap.

7 Empirical evidence

Protocols. Our primary confirmation is a frozen matched CIFAR-10 study with one CNN4-GN architecture, eight fresh paired seeds, a 4,710-update schedule, and all endpoints fixed before inspecting B. B appends a removable label-correlated ribbon while W receives the identical core with a zero ribbon. Controlled scope checks use a width-96 dense linear RNN grid ($\rho \in \{1, 2, 4, 8\}$, $\ell \in \{0, 2, 4, 8\}$, 8 seeds per cell) and width-32 tanh/GRU pairs at $(\rho, \ell) = (4, 2)$ (8 seeds). CDC is compared with ERM, Spectral Decoupling, and adapted shadow-target controls. Complete protocols and hyperparameters are in Appendix D.

7.1 A removable image cue causally suppresses core learning

All eight frozen CIFAR-10 pairs pass the causal certificate and its preregistered stricter at-hit subtype. Mean normalized signed deficit AUC is 0.489 (95% CI [0.413, 0.564]), compared with the frozen target 0.10, and the material deficit persists for 98.3% of training ([97.6%, 99.0%]). At the final checkpoint, B trails W by 19.9 percentage points on neutral inputs ($[-22.6, -17.2]$), 27.6 points under random cues ($[-28.7, -26.4]$), and 31.9 points under cue conflict ($[-33.0, -30.8]$), while gaining 16.8 points when the cue is label-consistent ($[15.8, 17.9]$).

With the original heads, the neutral B-W gap is -19.9 points. Equal fresh balanced linear heads reduce it to -10.8 points ($[-11.1, -10.4]$), a 9.1-point recovery ($[6.5, 11.8]$). Thus, part of the harm is readout underuse, while a substantial remainder is reduced linear accessibility under this probe; this is not evidence of information destruction.

7.2 Controlled recurrent regimes

Figure 3 and Table 2 present the audited dense-linear grid. At lags 0, 2, and 4, all 96 runs pass the causal certificate; the stricter at-hit subtype passes in 95/96 because one lag-0 run becomes negative only later. At lag 8, all 32 runs fail the learnability gate and remain indeterminate even though 27/32 have positive signed AUC. The mean deficit peaks at lag 2 (8.93) rather than decaying monotonically.

Figure 4 illustrates dynamic trajectories at the fixed nonlinear cell $(\rho, \text{lag}) = (4, 2)$.

- **Tanh RNN:** Passes the weak-only gate and at-hit certificate in 8/8 seeds: 3/8 transfer before suppressing, while 5/8 suppress throughout without a crossing.
- **GRU:** Exhibits direct-autograd drift crossings in 5/8 seeds and response crossings in 5/8 seeds. However, the weak-only model never reaches $\beta = 0.5$ ($\tau_W^* > 5.0$), resulting in **0/8 causal certificates**.

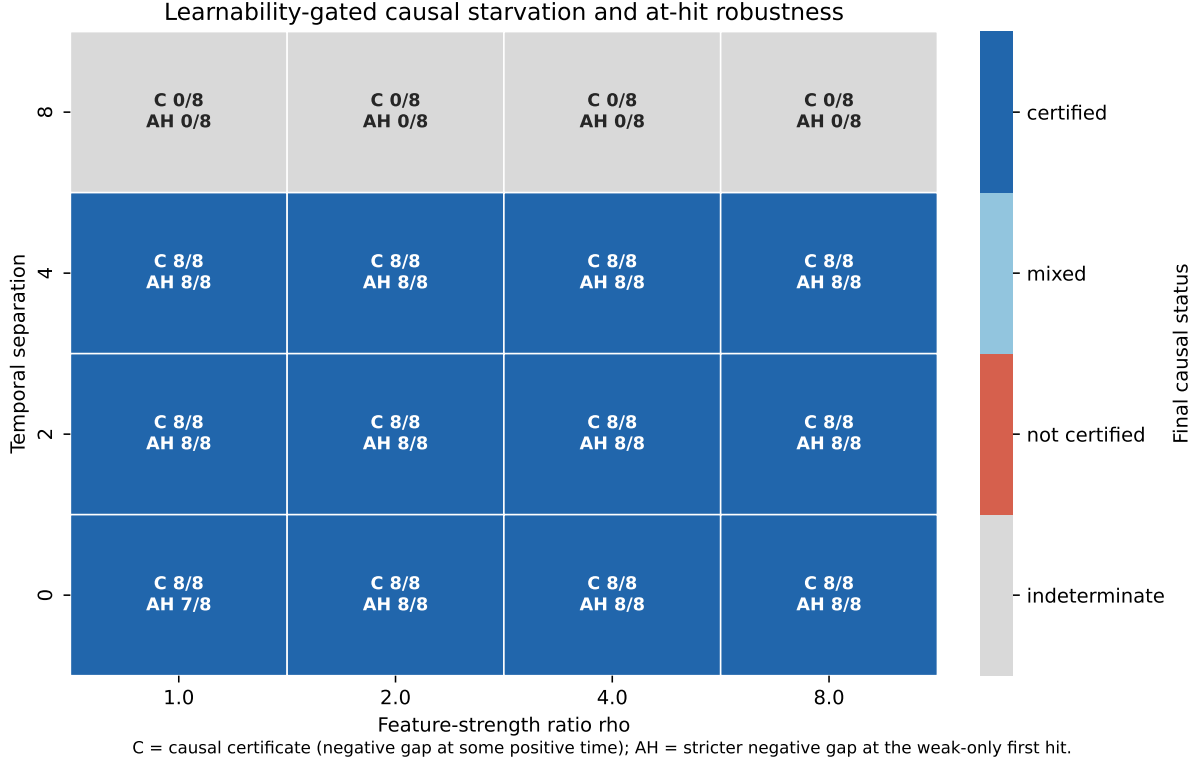


Figure 3: Final audited dense-linear causal grid. Cell color uses Definition 1; labels report causal-certificate (C) and stricter at-hit (AH) counts out of eight seeds. Lag-8 cells are indeterminate because the weak-only gate fails. Historical threshold-delay labels are not used.

This establishes empirically that crossing subtype and causal certificate are distinct: a run can be starved without crossing, while a crossing with a failed weak-only gate remains indeterminate.

7.3 Local mechanism control with CDC

Table E.3 summarizes the dense-linear mechanism control comparison across 8 paired seeds. All shadow-based interventions eliminate the weak AUC gap from ERM’s 10.52 to ≈ 0.00 . However, CDC uniquely preserves instantaneous strong drift (perturbation 4.77×10^{-7} , compared to $1.48 \times 10^{-1} - 3.75 \times 10^{-1}$ for alternatives). As predicted by theory, this local one-step guarantee does not imply maximum final strong response (1.042 vs. 1.295 for unconstrained rescue).

8 Limitations and conclusion

8.1 Scope and limitations

The empirical evidence is restricted to controlled synthetic grids and one preregistered CIFAR-10 intervention using an artificial ribbon cue and one architecture. It establishes a controlled existence result and diagnostic utility, not prevalence under natural shortcuts or a universal architecture ranking. The CIFAR experiment uses minibatch SGD and therefore does not verify the continuous-time tail-area theorem. The smooth-network theorem is conditional on certified uniform bounds, and its nonlinear tanh-CNN instance uses a constructed five-parameter initialization, a short dimensionless horizon, and a 10^{-7} response deficit; it does not certify the CIFAR CNN, nonlinear recurrent models, or typical random initializations. CDC requires an exact matched shadow model under full-batch gradient flow in a fixed parameter chart.

Table 2: Audited linear-grid outcomes. The sign-reversal row is an equivariance check, not a no-cue negative control.

Lag / intervention	Mean $\mathcal{A}_w$ (95% CI)	W gate	Causal	At-hit
0, $\rho = 1-8$	5.76 [5.01, 6.51]	32/32	32/32	31/32
2, $\rho = 1-8$	8.93 [8.29, 9.57]	32/32	32/32	32/32
4, $\rho = 1-8$	6.11 [5.31, 6.91]	32/32	32/32	32/32
8, $\rho = 1-8$	0.0015 [0.0010, 0.0020]	0/32	Indeterminate	Indeterminate
Sign reversal, (4, 4)	-4.04 [-4.50, -3.58]	8/8	2/8	0/8

8.2 Conclusion

Weak learning failure alone is not gradient starvation. The matched intervention first establishes whether the weak signal is learnable without the strong cue; exact finite-width dynamics and pathwise attribution then explain the response gap; certified parameter-region bounds can predict finite-horizon suppression in smooth networks, as demonstrated by the constructed tanh-CNN certificate. The CIFAR confirmation shows that this distinction has consequential behavioral and representational effects in one controlled image setting, while CDC illustrates local mechanism control.

Reproducibility statement

Complete source code, configurations, environment specifications, and content-addressed result manifests for the linear, recurrent, tangent, CDC, and frozen CIFAR-10 experiments are provided in the supplementary repository and documented in Appendix D.

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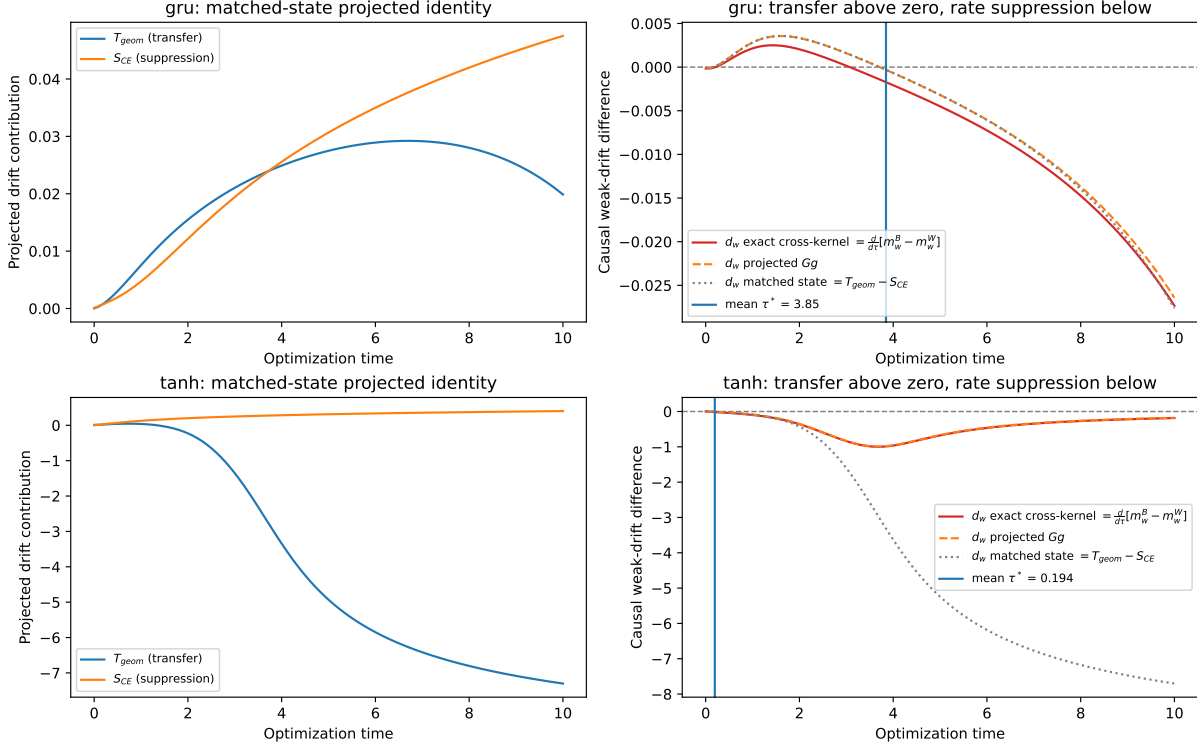


Figure 4: Paired nonlinear trajectories at $(\rho, \text{lag}) = (4, 2)$. Tanh (left) exhibits a certified causal starvation event. GRU (right) exhibits transient rate and response crossings but fails the weak-only gate, remaining causally indeterminate.

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AI use statement

Generative AI tools (OpenAI Codex/ChatGPT, Google Gemini, and Anthropic Claude) assisted with refining theoretical claims and proof arguments; designing experimental methodology; implementing, refactoring, and testing software; interpreting results; surveying and checking related-work candidates; and drafting, structuring, and editing the manuscript. The authors reviewed all AI-assisted work, independently checked the retained proofs, code, numerical artifacts, and citations, and take responsibility for the final text, claims, and artifacts.

A Full definitions and notation

A.1 Paired logistic gradient flow

Let $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$ be a dataset with binary labels $y_i \in \{-1, +1\}$ and inputs $x_i = (x_{s,i}, x_{w,i}, x_{bg,i}) \in \mathbb{R}^{d_s + d_w + d_{bg}}$. For condition $C \in \{B, W\}$, the parameter $\theta_C(\tau) \in \mathbb{R}^P$ evolves under full-batch gradient flow:

$$L_C(\theta) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-y_i f_C(x_i; \theta)} \right), \quad \frac{d\theta_C}{d\tau} = -\nabla L_C(\theta_C), \quad (\text{A.1})$$

where $f_B(x; \theta) = f(x; \theta)$ and $f_W(x; \theta) = f((0, x_w, x_{bg}); \theta)$. Both systems share initialization $\theta_B(0) = \theta_W(0) = \theta_0$.

A.2 Hitting time metrics and weak-AUC deficit

For any response functional $M(\tau)$ and target threshold $\beta > 0$:

$$\tau(\beta; H) = \inf\{\tau \in [0, H] : M(\tau) \geq \beta\}, \quad (\inf \emptyset = \infty). \quad (\text{A.2})$$

The paired weak-AUC deficit over horizon H is defined as:

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) d\tau = - \int_0^H \Delta(\tau) d\tau. \quad (\text{A.3})$$

B Additional theoretical results

B.1 Conditional tangent comparison

Freezing Θ and c_w at initialization defines $\dot{r} = \Theta_0 q(\bar{r})/n$ and $\dot{M}_w = c_{w,0}^\top q(\bar{r})/n$.

Theorem B.1 (Conditional tangent comparison). *Suppose on $[0, t]$ that $\|\Theta(u) - \Theta_0\| \leq B_\Theta$, $\|c_w(u) - c_{w,0}\| \leq B_c$, and $\|q(r)\| \leq V$. Let $a = \|\Theta_0\|/(4n)$ and $\psi_a(t) = (e^{at} - 1 - at)/a^2$. Then*

$$|M_w(t) - \bar{M}_w(t)| \leq \frac{B_c V}{n} t + \frac{\|c_{w,0}\| B_\Theta V}{4n^2} \psi_a(t). \quad (\text{B.1})$$

The bound is an exact conditional comparison, but this work does not establish nonvacuous B_Θ, B_c for trained tanh RNNs or GRUs. The empirical frozen-tangent study is therefore retained only as an event heuristic in Appendix E.

B.2 Closed projected flow under exact realization

Proposition B.3 (Closed projected flow). *If signed sample logits satisfy $r_i^C(\theta) = \sum_b z_{ib}^C M_b^C(\theta) + r_i^\perp(\theta)$, then:*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C + R_a^C, \quad (\text{B.2})$$

where $G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle$, $g_b^C = \frac{1}{n} \sum_i z_{ib}^C \sigma(-r_i^C)$, and $R_a^C = \frac{1}{n} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle$. When realization is exact ($r_i^\perp \equiv 0$), $R_a^C \equiv 0$ and the response dynamics close autonomously.

B.3 Log drift-ratio identity

Theorem B.2 (Log drift-ratio identity). *For rank-one positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, write $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$ where $H_B = \rho G_{ws}^B + G_{ww}^B$ and $H_W = G_{ww}^W$. Define $\Psi = \log(F_w^B/F_w^W)$. Then $\text{sign}(d) = \text{sign}(\Psi)$, and:*

$$\dot{\Psi} = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \frac{d}{d\tau} \log \left(\frac{H_B}{H_W} \right), \quad (\text{B.3})$$

where $Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0$.

B.4 Every-width six-scalar solvable zero-lag limit

Theorem B.3 (Every-width six-scalar closure at zero lag). *At zero temporal lag and zero background noise in the dense linear RNN, the recurrent weight matrix W is loss-invisible ($\nabla_W L = 0$). With $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$, the dynamics close at every width N :*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad \dot{w} = 2 \sum_b g_b M_b. \quad (\text{B.4})$$

C Proofs

C.1 Proof of the exact finite-width response-logit dynamics

Under full-batch logistic gradient flow, $\dot{\theta} = -\nabla L(\theta) = \frac{1}{n} \sum_{i=1}^n \sigma(-r_i) \nabla r_i(\theta) = \frac{1}{n} J(\theta)^\top q(r)$. By chain rule, for any response $M_w(\theta)$: $\dot{M}_w = \langle \nabla M_w, \dot{\theta} \rangle = \frac{1}{n} \langle \nabla M_w, J^\top q(r) \rangle = \frac{1}{n} (J \nabla M_w)^\top q(r) = \frac{1}{n} c_w^\top q(r)$. Similarly, $\dot{r}_i = \langle \nabla r_i, \dot{\theta} \rangle = \frac{1}{n} \sum_j \langle \nabla r_i, \nabla r_j \rangle q_j = \frac{1}{n} (\Theta q(r))_i$. $\square$

C.2 Proof of cumulative drift balance

Since $\Delta(0) = 0$ and $\Delta(\tau)$ is absolutely continuous, $\Delta(t) = \int_0^t \dot{\Delta}(u) du = \int_0^t d(u) du$. Writing $d(u) = [d(u)]_+ - [-d(u)]_+$ gives $\Delta(t) = \int_0^t [d(u)]_+ du - \int_0^t [-d(u)]_+ du = P(t) - N(t)$. Thus $\Delta(t) < 0 \iff P(t) - N(t) < 0 \iff N(t) > P(t)$. $\square$

C.3 Proof of the conditional tangent comparison

Let $e_r(t) = r(t) - \bar{r}(t)$ and $e_M(t) = M(t) - \bar{M}(t)$. Differentiating: $\dot{e}_r = \frac{1}{n} [\Theta(t)q(r(t)) - \Theta_0 q(\bar{r}(t))] = \frac{1}{n} [\Theta(t) - \Theta_0]q(r(t)) + \frac{1}{n} \Theta_0 [q(r(t)) - q(\bar{r}(t))]$. Since σ' is bounded by $1/4$, $\|q(r) - q(\bar{r})\| \leq \frac{1}{4} \|e_r\|$. Applying Grönwall's inequality with $a = \|\Theta_0\|/(4n)$ yields the bounds in Eqs. (B.1). $\square$

C.4 Proof of the realized-initialization rate-gap criterion

At $\tau = 0$, $\theta_B(0) = \theta_W(0)$. Using the dense linear block gradients with $W(0), B(0), c(0)$, $G_{ws}(0) = b_s^\top W^\ell b_w = K$ and $G_{ww}(0) = A$. Evaluating the initial weak drift in each condition: $\dot{M}_w^B(0) = s_B(\rho K + A)$ and $\dot{M}_w^W(0) = s_W A$. Subtracting gives $d_0 = s_B(\rho K + A) - s_W A$. Since $\rho s_B > 0$, rearrangement yields Eq. (11). Shared initialization gives $\Delta(0) = 0$ and differentiability gives $\Delta(t)/t \rightarrow \Delta'(0) = d_0$ as $t \downarrow 0$. Therefore a strict negative (positive) d_0 implies strict local outcome suppression (transfer) on some sufficiently small positive interval. The argument is existential and supplies no numerical endpoint. $\square$

C.5 Proof of the parameter-tube prediction theorem

Because $L_C, M_w \in C^2$, the paired field is locally Lipschitz, giving local existence and uniqueness. Before exit from U_R , for $0 \leq u \leq h$,

$$\|X(t_w + u) - \bar{X}_w\| \leq r_w + \int_0^u \|V(X(t_w + s))\| ds \leq r_w + Su \leq R. \quad (\text{C.1})$$

Thus no first exit can occur before H ; bounded velocity on the compact ball also prevents a finite maximal endpoint there. Along the contained trajectory, $d' = j \leq -\lambda$, so $d(t_w + u) \leq d_+ - \lambda u$. Integrating $\Delta' = d$ and $\dot{M}_w^W \geq \nu$ gives

$$\Delta(t_w + u) \leq D_+ + d_+ u - \frac{\lambda u^2}{2} = Q(u), \quad M_w^W(t_w + u) \geq m + \nu u. \quad (\text{C.2})$$

For $u \geq u_a$, $Q'(u) = d_+ - \lambda u \leq 0$; hence $Q(u) \leq Q(u_a) < 0$ through H . The weak-response lower bound reaches β by H when $m + \nu h \geq \beta$; continuity and $M_w^W(0) < \beta$ make its first hit strictly positive. This proves the broad causal certificate without requiring suppression specifically at the hit. For the converse, $|d'| \leq J$ gives $d(t_w + u) \geq d_- - Ju > 0$; integrating from $\Delta(t_w) \geq D_- \geq 0$ excludes a later negative gap. $\square$

C.6 Certificate for the nonlinear tanh-CNN construction

For Eq. (9), oddness of tanh makes the two signed training logits within each condition identical and gives

$$\begin{aligned} r_B(\theta) &= \frac{c}{2}[\tanh(s_0 + s_1 + w_1) + \tanh(s_0 + w_0 + w_1)], \\ r_W(\theta) = M_w(\theta) &= \frac{c}{2}[\tanh(w_1) + \tanh(w_0 + w_1)]. \end{aligned} \quad (\text{C.3})$$

This is the exact nonlinear trained graph, not a tangent replacement. It is C^∞ in all five trainable parameters. At $t_w = 1/10000$, the supplied outward-rounded interval certificate encloses the warm-up state within $r_w \leq 1/10^6$ of its midpoint and verifies uniformly on the paired ball

$$\begin{aligned} R &= \frac{1}{200}, & S &\leq \frac{5}{4}, & r_w &\leq \frac{1}{1,000,000}, \\ -\frac{3}{10} &\leq j \leq -\frac{3}{32}, & \dot{M}_w^W &\geq \frac{1}{2}, \\ D_+ &\leq \frac{3}{50,000,000}, & d_+ &\leq \frac{1}{50,000}, & m &\geq \frac{3}{50,000}. \end{aligned} \quad (\text{C.4})$$

With $h = 39/10000$ and the prespecified $u_a = 29/10000$, exact rational arithmetic yields

$$r_w + Sh \leq \frac{1}{10^6} + \frac{5}{4} \frac{39}{10^4} = \frac{1219}{250000} < \frac{1}{200}, \quad (\text{C.5})$$

$$Q(u_a) \leq \frac{3}{50,000,000} + \frac{1}{50,000} \frac{29}{10^4} - \frac{3}{64} \left(\frac{29}{10^4} \right)^2 = -\frac{8839}{32,000,000,000} < -10^{-7}, \quad (\text{C.6})$$

$$Q'(u_a) \leq \frac{1}{50,000} - \frac{3}{32} \frac{29}{10^4} = -\frac{403}{1,600,000} < 0, \quad (\text{C.7})$$

$$m + \nu h \geq \frac{3}{50,000} + \frac{1}{2} \frac{39}{10^4} = \frac{201}{100000} > \frac{3}{2000}. \quad (\text{C.8})$$

Theorem 1 therefore proves the future deficit and nondegenerate W gate, with $t_w + (\beta - m)/\nu \leq 149/50000$. A separate interval enclosure on the warm-up verifies $d(0) \geq 3/100000$ and $j \geq -1/4$, hence

$$\Delta(t) \geq \frac{3t}{100000} - \frac{t^2}{8} > 0 \quad (0 < t \leq 1/10000), \quad (\text{C.9})$$

establishing genuine transfer before suppression.

For full-batch gradient descent, interval induction with $I_{k+1} = I_k + \eta V(I_k)$ encloses the exact update from each certified box. At $\eta = 10^{-4}$, the certificate boxes through update 40 verify Eq. (10)'s discrete deficit and gate statements. For the control initialization, the corresponding ball verifies $|j| \leq 1$, a positive cutoff gap, and $d_- - (H - t_w) > 0$; the safe-transfer part of Theorem 1 excludes suppression through $H = 1/500$. $\square$

C.7 Proof of the CDC local guarantee

For (1): $a \cdot v' = a \cdot (v + \alpha q) = a \cdot v + \alpha(a \cdot q) = a \cdot v$, since $a \cdot q = a \cdot (p - \frac{p \cdot a}{\|a\|^2} a) = 0$. For (2): The optimization problem is $\min_d \frac{1}{2} \|d\|^2$ subject to $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$. Forming the Lagrangian $L(d, \lambda_1, \lambda_2) = \frac{1}{2} \|d\|^2 - \lambda_1(a \cdot d) - \lambda_2(p \cdot d - \delta)$, stationarity gives $d^* = \lambda_1 a + \lambda_2 p$. Orthogonality $a \cdot d^* = 0$ forces $\lambda_1 = -\lambda_2 \frac{a \cdot p}{\|a\|^2}$, so $d^* = \lambda_2 q$. Enforcing $p \cdot d^* = \delta$ yields $\lambda_2 = \delta / \|q\|^2 = \alpha$, proving uniqueness. For (3): By Taylor expansion with Lipschitz gradient ∇m_s , $|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \eta |a \cdot (v' - v)| + \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2) = \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2)$. $\square$

D Experimental details and complete hyperparameters

D.1 Frozen CIFAR-10 confirmation details

The confirmatory source commit is 0726aea12947a30eb60ae32fc9c1eb0a7a122660; the frozen configuration and DGX environment have SHA-256 digests 8f662f9982f8445b and 1c1e2d126093f449, respectively (prefixes shown). Runs use CIFAR-10 archive SHA-256 6d958be074577803. The split contains 40,000 training,

Table D.1: Complete experimental hyperparameters across all study designs.

Hyperparameter	Linear Grid	Nonlinear Fixed-Cell	Tangent Held-Out	CDC Ablation
Architecture	Dense Linear RNN	Tanh RNN / GRU	Tanh RNN / GRU	Dense Linear RNN
Hidden width N	96	32	32	96
Recurrent Gain g	0.5	0.8	0.8	0.5
Input scaling σ_B	$1/\sqrt{N}$	$1/\sqrt{N}$	$1/\sqrt{N}$	$1/\sqrt{N}$
Sequence length T	10	10	10	10
Optimizer	Full-batch GD	Full-batch GD	Full-batch GD	Full-batch GD
Learning rate η	0.1	0.05	0.05	0.1
Horizon H	5.0 (50 steps)	5.0 (100 steps)	5.0 (100 steps)	5.0 (50 steps)
Target threshold β	0.25	0.5	0.5	0.25
Replication	8 seeds $\times$ 16 cells	8 seeds $\times$ 2 archs	4 data $\times$ 8 model seeds	8 paired seeds
Hardware	Apple M-series / CUDA	Apple M-series / CUDA	Apple M-series / CUDA	Apple M-series / CUDA

5,000 validation, 2,500 probe-fit, and 2,500 diagnostic-evaluation images. Full hashes, raw traces, and the independent audit are included in the supplementary artifacts.

Input construction and augmentation. Each 32×32 RGB core is scaled to $[-1, 1]$ and extended to a $3 \times 40 \times 32$ tensor by appending an eight-row ribbon. B receives one of ten 4×4 nonconstant Sylvester–Hadamard codes at a seed-fixed random location (row 0–4, column 0–28), with label-code reliability 0.9 and amplitude 0.5; unreliable examples receive a uniformly selected wrong code. W receives the identical augmented core and an exactly zero ribbon after normalization. Training augmentation is reflection padding by four pixels, a seeded random 32×32 crop, and a horizontal flip with probability 1/2. B and W share source-image splits, crop/flip randomness, minibatch order, initialization, and optimizer settings.

Architecture, optimization, and response. CNN4-GN has four 3×3 convolution–GroupNorm(8)–ReLU– 2×2 max-pool blocks with channels (32, 64, 128, 128), adaptive average pooling, and a ten-class linear head. We use SGD with learning rate 0.05, momentum 0.9, weight decay 5×10^{-4} , batch size 256, and cosine decay over 4,710 updates, logging every 157 updates. On the fixed common neutral diagnostic set, class-balanced Brier skill is

$$1 - \frac{\frac{1}{10} \sum_{k=1}^{10} \mathbb{E}[\|p - e_k\|^2 \mid y = k]}{1 - 1/10}, \quad (\text{D.1})$$

so uniform predictions score zero and perfect predictions score one. The frozen gate uses $\beta = 0.5$. E4-lite freezes the final encoders and fits zero-initialized balanced linear softmax heads on the common probe-fit split, selecting $\ell_2 \in \{0, 10^{-3}, 10^{-2}\}$ by validation cross-entropy with at most 100 iterations before one held-out evaluation.

Table D.2: Per-seed frozen CIFAR-10 outcomes. Duration is the longest material core-response deficit as a fraction of the 4,710-update horizon. Accuracy gaps are B–W on neutral inputs in percentage points.

Seed	W first hit	Signed deficit AUC	Duration	Original-head gap	Fresh-head gap
33001	942	0.351	0.986	−16.44	−10.04
33002	1256	0.573	0.990	−23.84	−11.28
33003	785	0.495	0.974	−18.80	−10.60
33004	1099	0.616	0.989	−24.48	−10.72
33005	942	0.442	0.968	−18.48	−11.48
33006	942	0.495	0.986	−21.00	−10.40
33007	1256	0.543	0.978	−20.64	−10.76
33008	1099	0.393	0.993	−15.48	−10.76

E Supplementary experiments and ablations

E.1 Width trends and finite-network closure validation

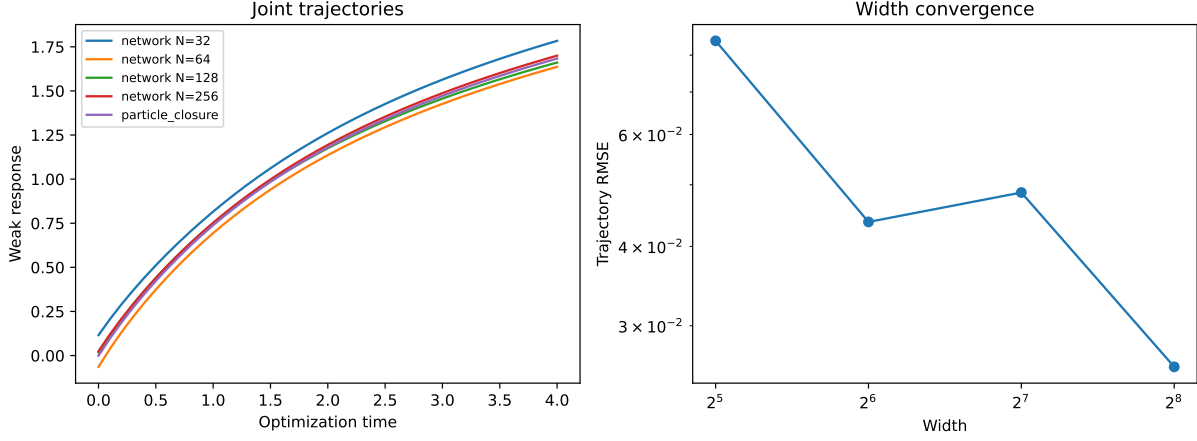


Figure E.1: Finite-width agreement with held-out empirical closure reference across widths $N \in \{32, 64, 128, 256\}$.

E.2 Frozen-tangent event heuristic

The sealed held-out study contains 64 crossed records (four unseen data seeds by eight unseen model seeds for each architecture). The frozen surrogate classifies response-crossing events in 64/64 records and drift crossings in 60/64, but it does not integrate the weak-only trained trajectory and therefore has zero coverage for the learnability-gated causal endpoint. This is an event-classification result, not a calibrated dynamical predictor or evidence of tangent-kernel stability.

Table E.1: Restricted held-out frozen-tangent evaluation. “Coverage” denotes whether the surrogate can produce the endpoint, not predictive accuracy.

Endpoint	Correct event classifications	Endpoint coverage
Drift crossing	60/64	64/64
Response crossing	64/64	64/64
Causal starvation	—	0/64

E.3 Semi-real generated-cue study

Table E.2: Semi-real generated-cue study on MNIST (3-vs-8) and FashionMNIST (0-vs-6) across 64 crossed records.

Dataset	Weak-Only Learnable	Causal Certificate	Mean Weak-AUC Gap (95% CI)
MNIST 3-vs-8	0/32	0/32	−0.00124 [−0.00344, 0.00102]
FashionMNIST 0-vs-6	0/32	0/32	−0.00011 [−0.00413, 0.00318]

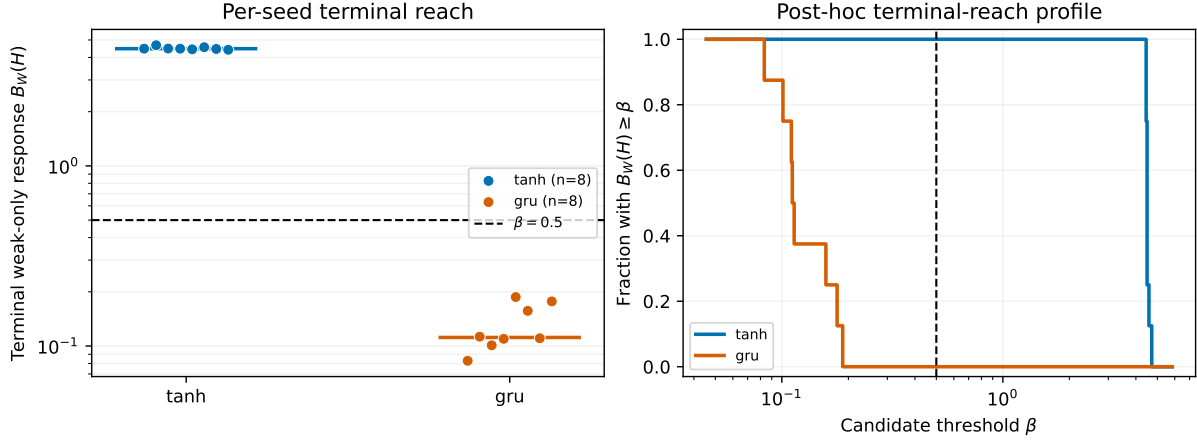


Figure E.2: Post-hoc weak-only terminal response distributions and threshold sensitivity profiles across seeds.

E.4 Threshold sensitivity and β -sweeps

E.5 Full 8-method mitigation ablation

Table E.3: Dense CDC ablation across 8 paired seeds. Shadow methods access a matched weak-only control.

Method	Shadow	Weak AUC Gap	Strong Drift Perturbation	Final M_s
ERM	No	10.516	—	1.364
Spectral Decoupling	No	9.023	—	0.829
CDC (Ours)	Yes	−0.0005	4.77×10^{-7}	1.042
Unconstrained Rescue	Yes	−0.0007	1.48×10^{-1}	1.295
Adapted Bloop-style	Yes	0.0002	3.75×10^{-1}	0.370
Adapted PCGrad-style	Yes	−0.0007	1.48×10^{-1}	1.295

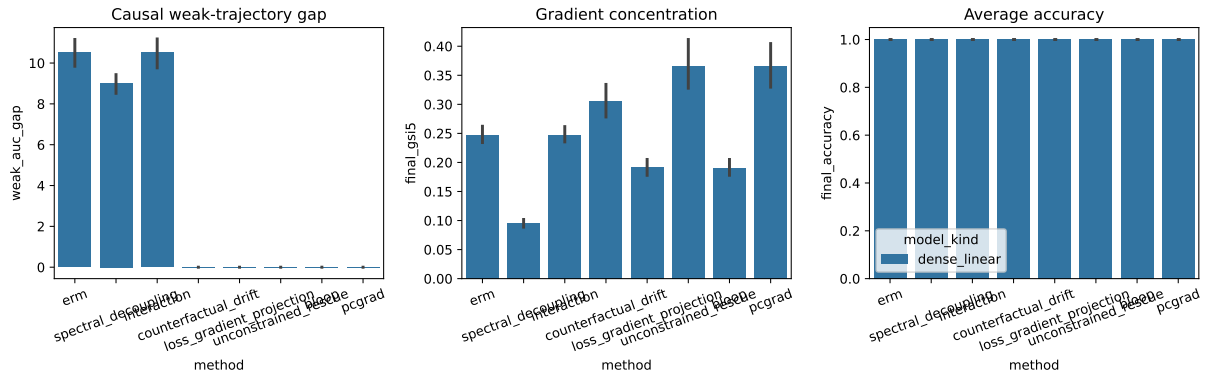


Figure E.3: Full eight-method dense-linear mitigation ablation comparing CDC with ERM, Spectral Decoupling, PCGrad, Bloop, and unconstrained rescue.